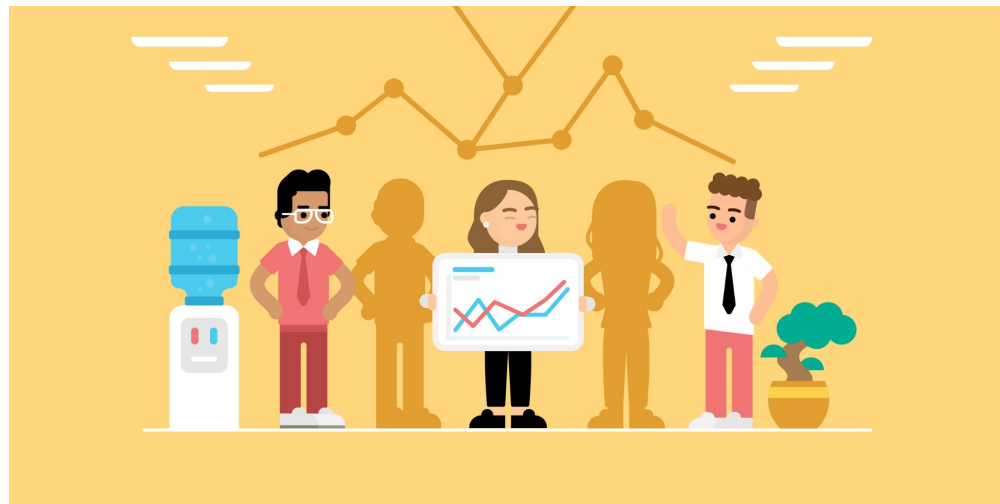


quantitative methods in sociolinguistics

Yvette Yi-Chi Wu



studying linguistic variation: a review

- **orderly heterogeneity** (Weinreich et al. 1968): speaker's choice between variable linguistic forms are systematically constrained
- language change is **perpetual and ongoing** at all times (Tagliamonte 2006), although this is often ignored in generative linguistics
- sociolinguistic research on **non-English languages**, particularly sociolinguistic fieldwork and documentation, is still very much lacking!
- in recent sociolinguistics, emphasis has been moving towards the idea that linguistic variation is **created to signal social stratification** (Eckert 2012)
- **synchronic variation** is often a reflection of **diachronic change** (Labov 1994)

quantitative modelling

- strength of association between a **linguistic variable** and a **contextual feature**
 - some linguistic variables: code-switching, vowel quality, voice quality, lexical use, etc.
 - some contextual features: linguistic environment, audience, social phenomena, etc.
- **principle of multiple causes**: it is unlikely that any single contextual factor can explain the variability observed in natural data
- harder to address **variation in syntax, discourse, and code-switching** (rather than phonology), as it is more difficult to identify what an instance of a variable is



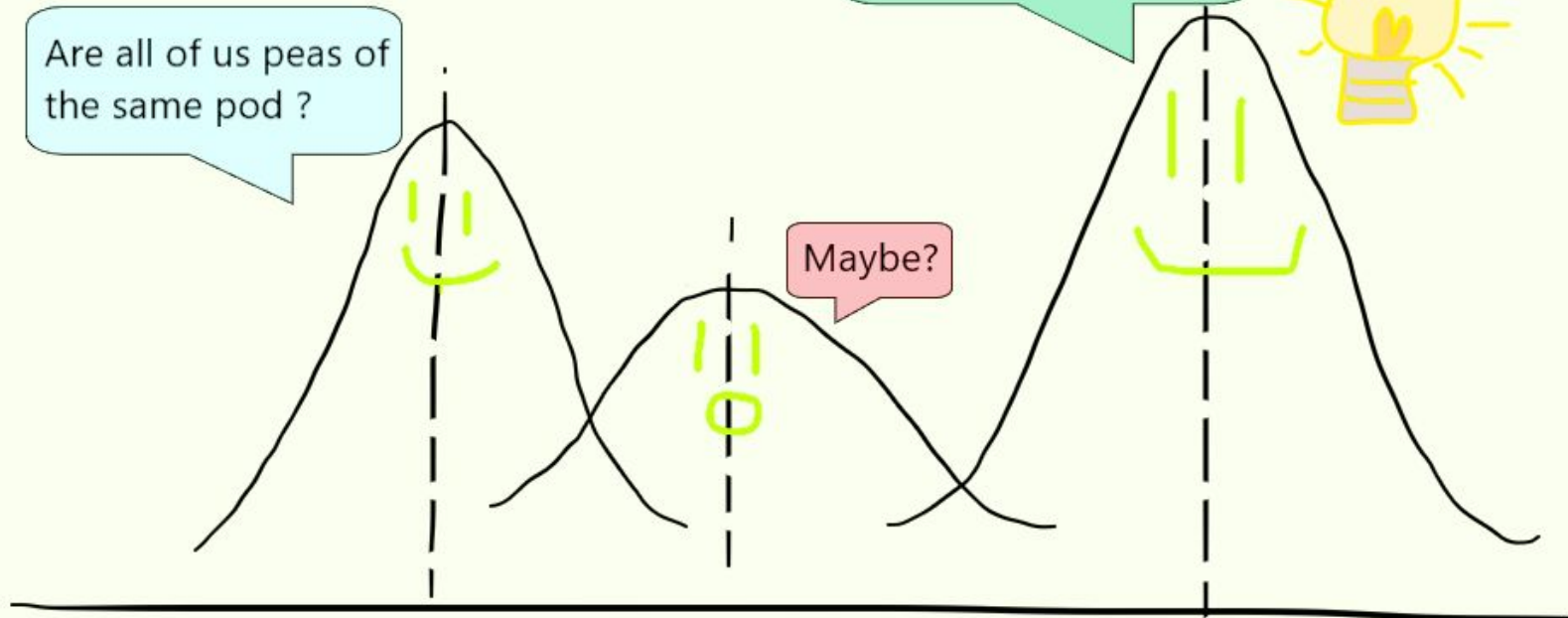
quantitative modelling: some terms

- **statistical significance**: whether a factor contributes to the variation observed, as opposed to by chance, assuming a certain probability distribution (< 0.05)
- for **continuous or quantitative** variables:
 - **t-test**: compares the means of two groups (good for small sample sizes)
 - **ANOVA (analysis of variance)**: compares the means of three or more groups
 - **linear regression**: examines the relationship between continuous independent and dependent variables (can have multiple independent variables)
- for **categorical or qualitative** variables:
 - **chi-square**: determines if there is significant correlation between two categorical variables (good for large sample sizes)

Are all of us peas of the same pod ?

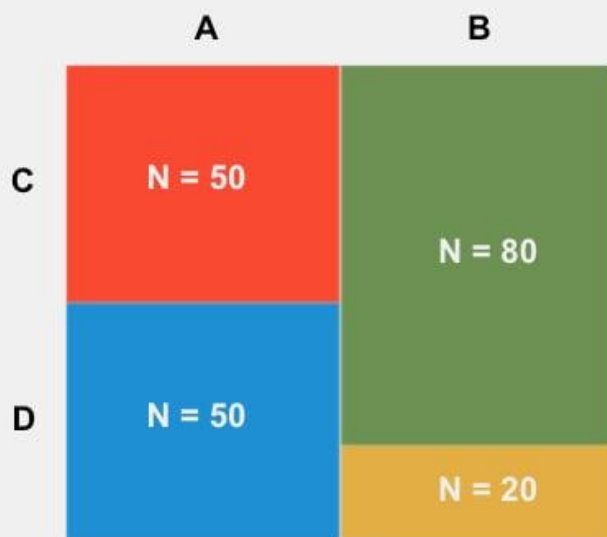
I have an idea! Let's use ANOVA and find out!

Maybe?



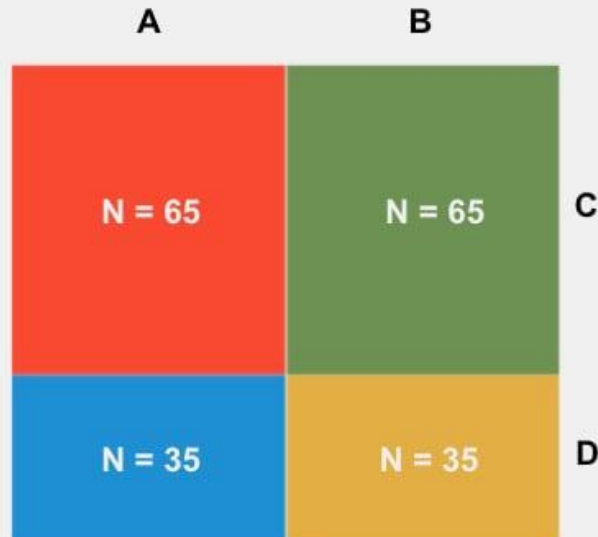
Chi-Square Test of Independence

Observed

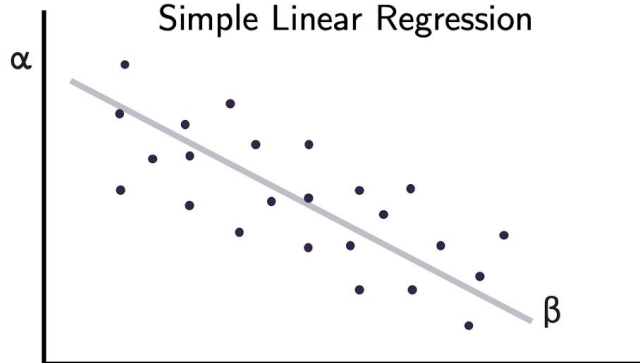


?

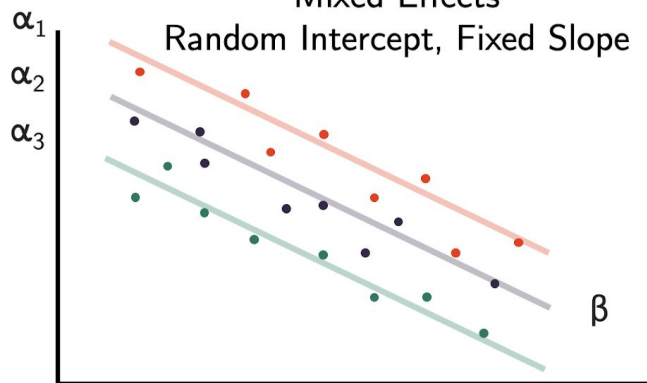
Expected



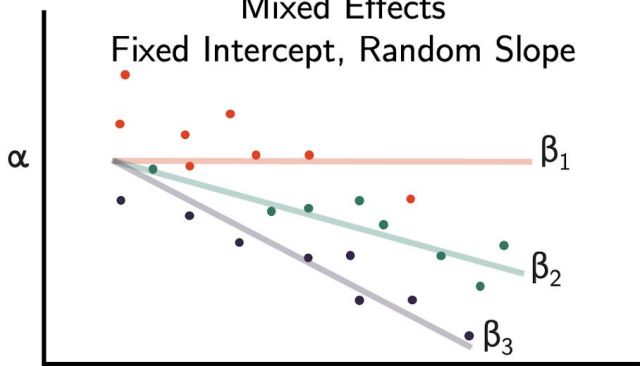
Fixed Effects
Simple Linear Regression



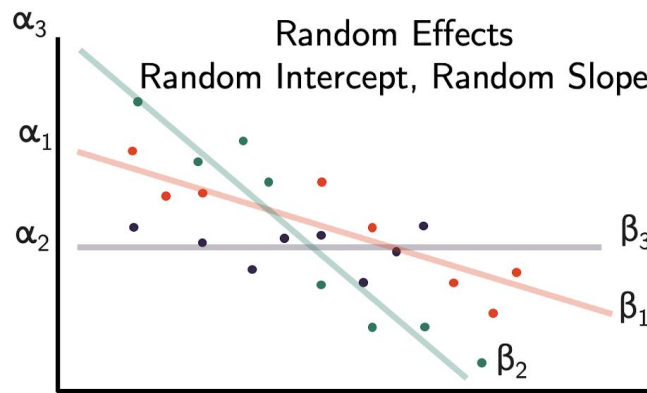
Mixed Effects
Random Intercept, Fixed Slope



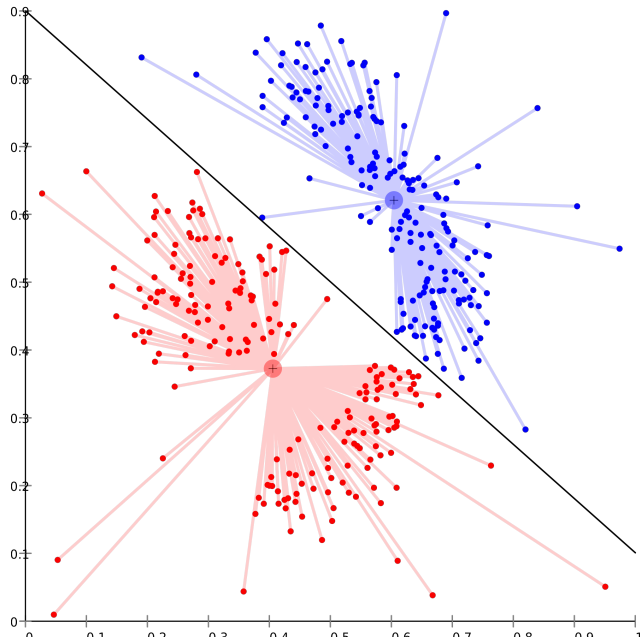
Mixed Effects
Fixed Intercept, Random Slope



Random Effects
Random Intercept, Random Slope

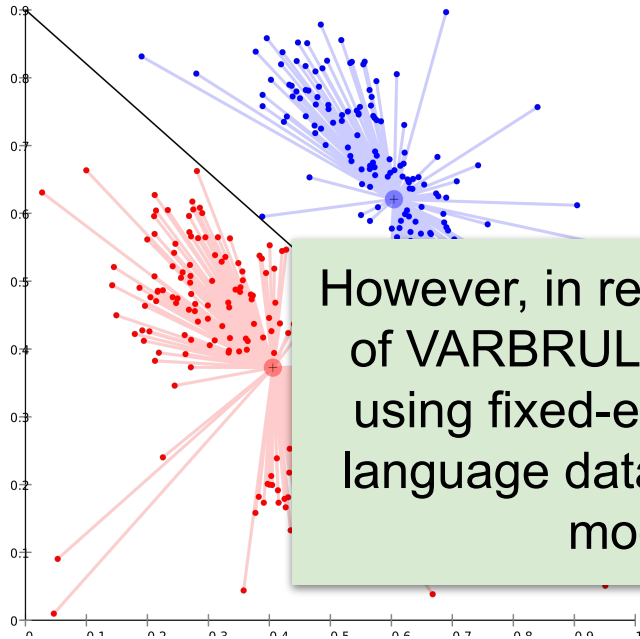


multivariate analysis in sociolinguistics



- VARBRUL (variable rules analysis) in GoldVarb has been common in the past
- designed for sociolinguistic variation:
 - too many variables for some other analyses
 - empty cells and imbalanced distribution due to naturalistic (vs. controlled) language data
- VARBRUL estimates the probabilities of each contextual factor specified, and measures the relative influence of each factor on the occurrence of the linguistic variable

multivariate analysis in sociolinguistics



- VARBRUL (variable rules analysis) in GoldVarb has been common in the past
- designed for sociolinguistic variation:

However, in recent years there has been critique of VARBRUL and the field has turned against using fixed-effects models to address natural language data. Now, mixed-effects regression models are more popular.

...other analyses
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